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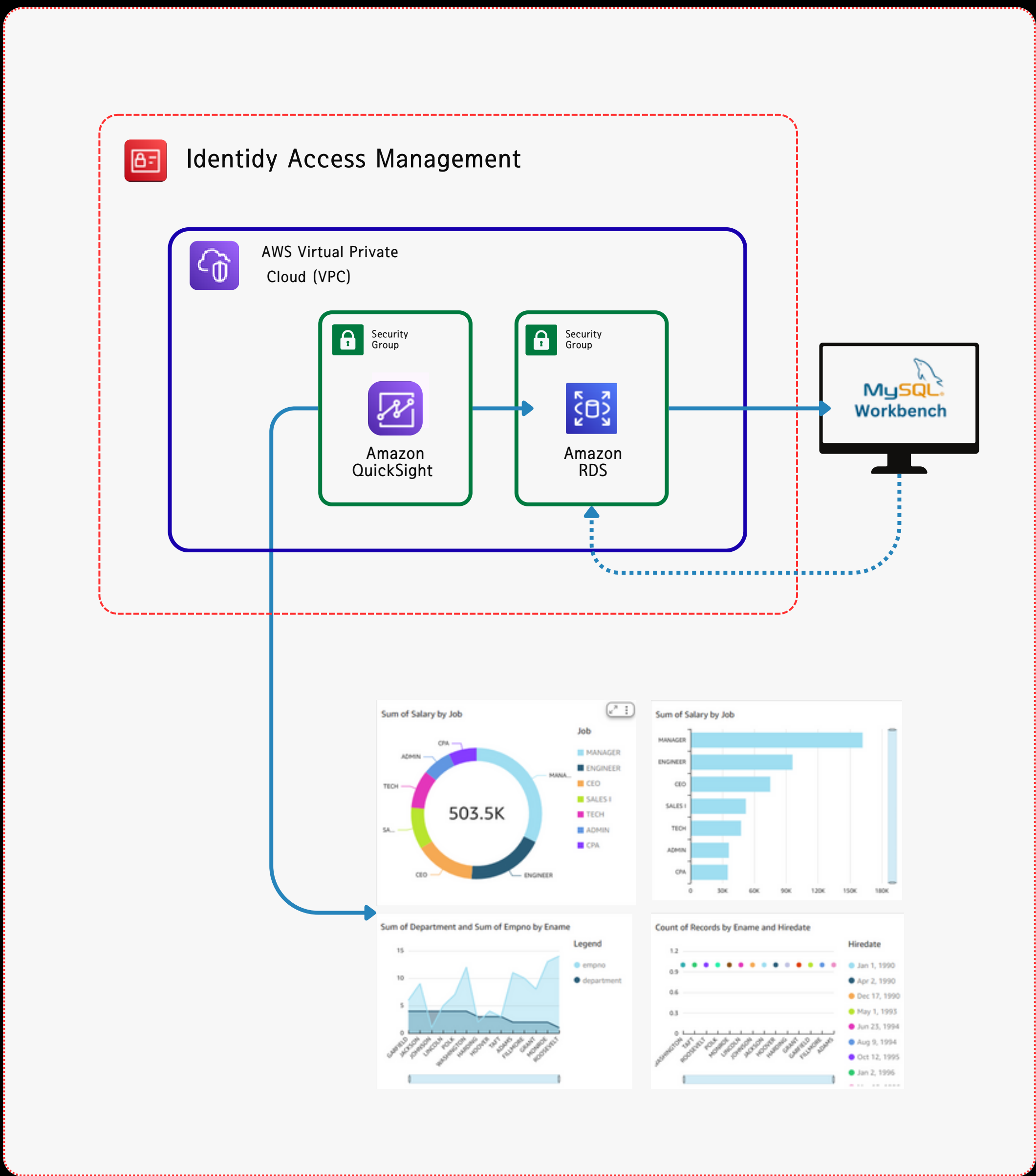
Community Lead

DATABASE PRO IN THE MAKING

Cloud Project:

Visualizing Relational Database using Amazon RDS and QuickSight



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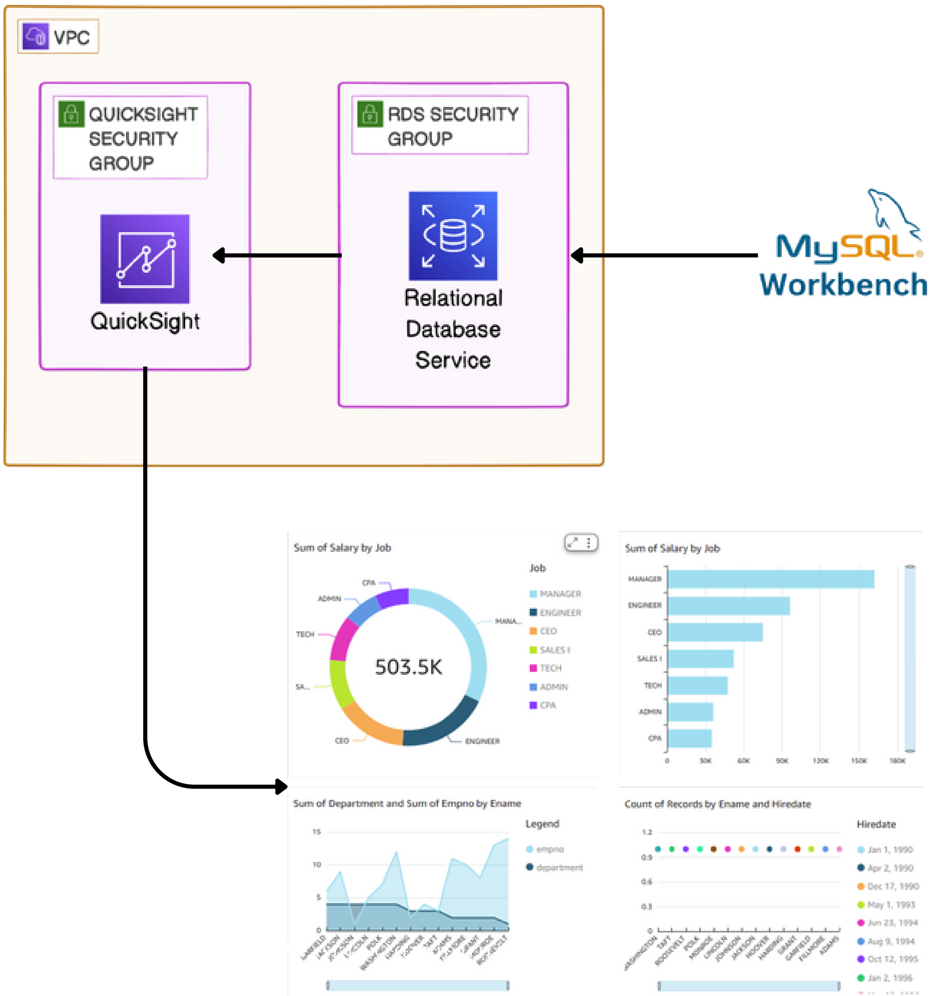
1 INTRODUCTION

This project demonstrates how to build, secure, and visualize a relational database using AWS services and third-party tools. The main objective is to set up a relational database in Amazon RDS, connect it with MySQL Workbench for management, and then use Amazon QuickSight for data visualization and reporting. Security and access are managed through IAM, Security Groups, and VPC configurations to ensure a reliable and protected environment.



TOOLS AND SERVICES

Amazon RDS (Relational Database Service)
- Hosts the MySQL database in a managed environment.
- Automates tasks such as patching, backups, and scaling.

Amazon VPC (Virtual Private Cloud)
- Defines the isolated network environment for the database.
- Ensures the RDS instance is deployed in a private subnet.

Security Groups
- Acts as a firewall for controlling inbound and outbound traffic.
- Restricts database access only to approved IPs

IAM (Identity and Access Management)
- Manages user roles and policies for AWS resources.
- Provides least-privilege access -
- Ensures only authorized users can manage or query the database.

MySQL Workbench
- Local client for database administration and schema management.
- Used for testing queries, designing tables, and importing/exporting data.

Amazon QuickSight
- Business intelligence (BI) service for creating interactive dashboards.
- Directly connects to Amazon RDS to visualize relational data.

2 UNDERSTANDING RELATIONAL DATABASES

WHAT IS AMAZON RDS?

Amazon RDS (Relational Database Service) is a managed relational database service offered by Amazon Web Services (AWS). It is a web service that simplifies the setup, operation, and scaling of a relational database in the cloud.

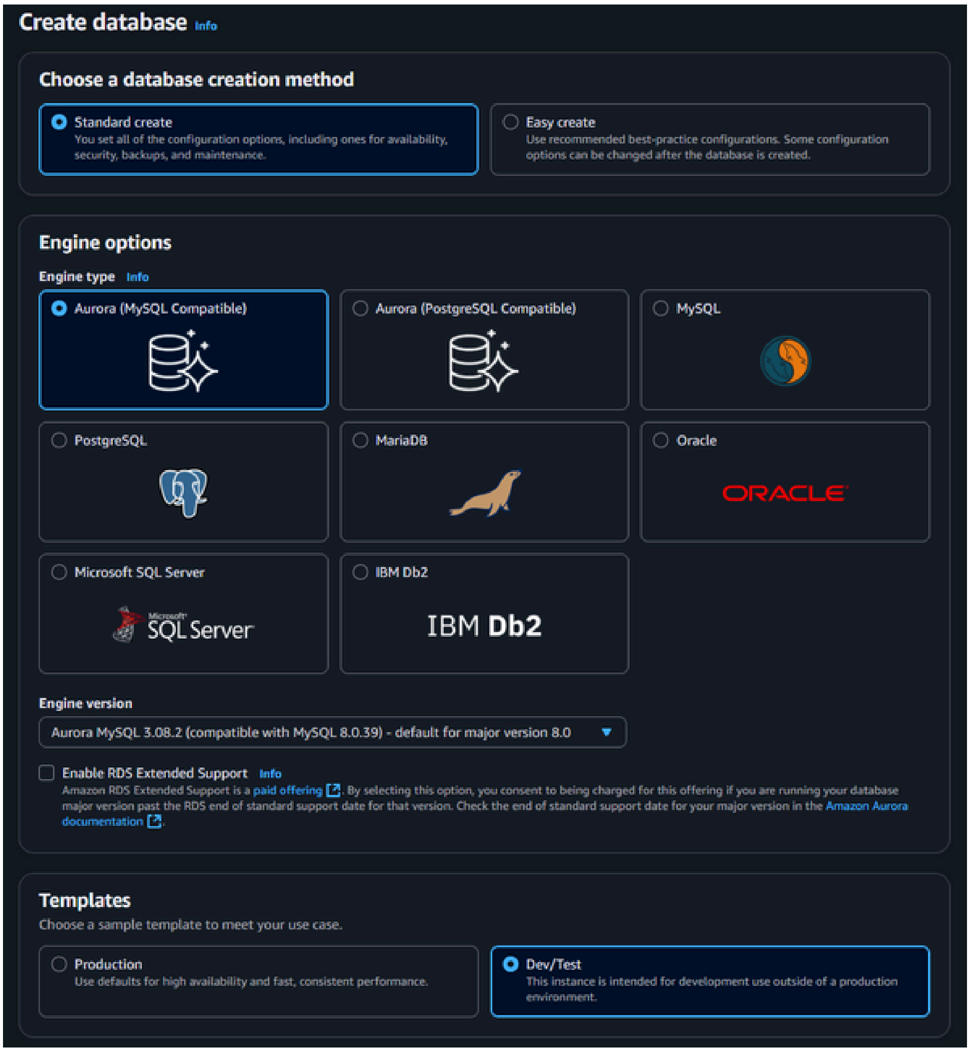
HOW I USED AMAZON RDS IN THIS PROJECT

In today's project, I used Amazon RDS to provision and manage a secure, scalable relational database in the cloud. It automated administrative tasks like patching, backups, and high availability, allowing me to focus on data visualization.

This project took about four hours to complete. The most time-consuming part was troubleshooting the VPC connection and IAM permissions. The actual chart creation in QuickSight was straightforward.

CREATING RELATIONAL DATABASE

I created my relational database by using Aurora and the RDS console, which allowed me to quickly provision a highly scalable and fault-tolerant database cluster. Aurora's compatibility with MySQL made it easier to connect with standard tools, while the RDS console provided options for automated backups, monitoring, and parameter configuration. To ensure secure deployment, the cluster was placed inside a VPC with private subnets, and Security Groups were applied to control inbound traffic, limiting access only to approved sources.



UNDERSTANDING RELATIONAL DATABASES

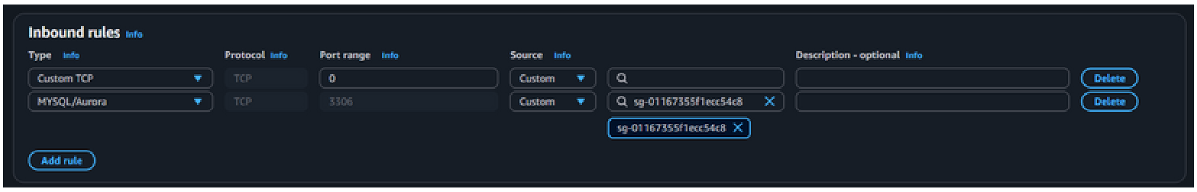
A relational database stores information in a structured format using tables made up of rows and columns. Each table represents a specific type of data, and predefined relationships between these tables allow connections across different data points. This model ensures data consistency, reduces redundancy, and makes it easier to organize and query large amounts of information efficiently.

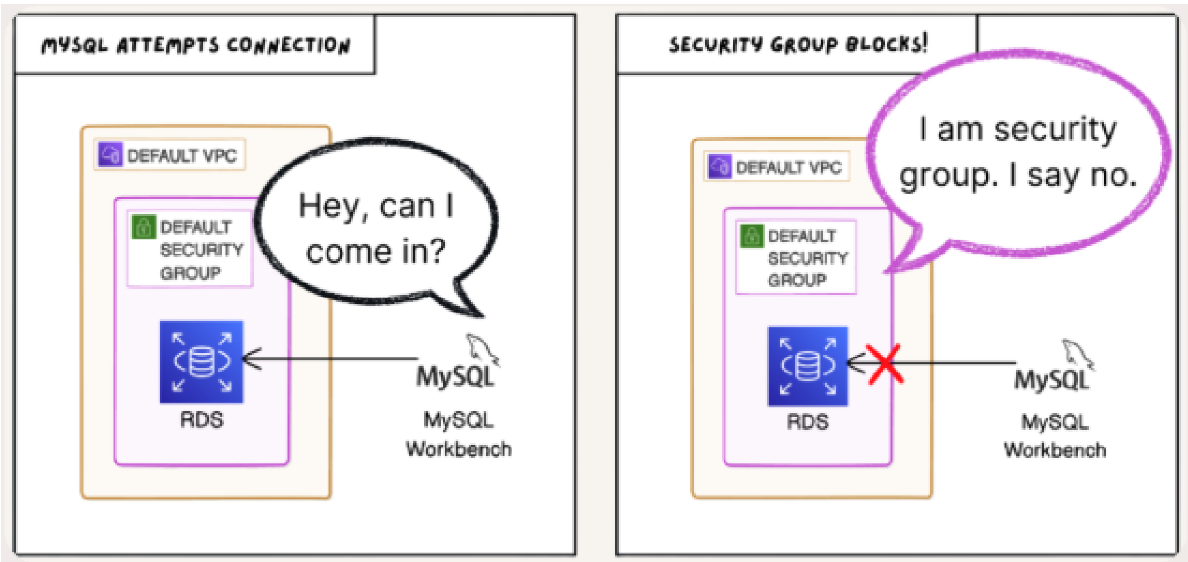
MYSQL VS SQL

SQL, or Structured Query Language, is the standard programming language used to manage and manipulate relational databases. MySQL, on the other hand, is a Relational Database Management System (RDBMS) that implements SQL to handle database operations such as creating tables, inserting records, and running queries. In short, SQL is the language, while MySQL is the software that uses that language to manage databases.

POPULATING MY RDS INSTANCE

First, I made my RDS instance publicly accessible so it could be reached from my local machine and third-party tools. I then updated the default security group, since it initially blocked inbound and outbound traffic. By configuring the security group with specific rules to allow access from trusted IP addresses, I ensured that external software such as MySQL Workbench could securely connect to the RDS instance for data population and management.

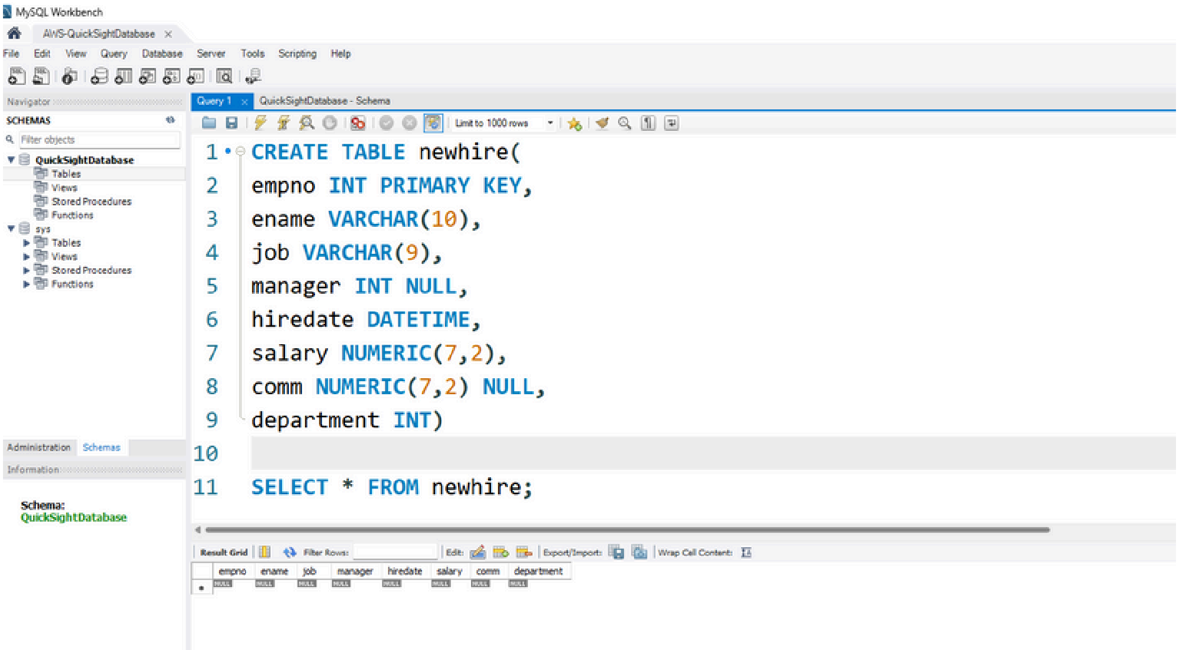




3 MYSQL WORKBENCH

USING MYSQL WORKBENCH

To populate my database, I manually executed SQL queries to first define the structure and then insert initial records. This involved creating tables with the appropriate columns, data types, and relationships, followed by adding sample data to ensure the schema was functional and ready for integration with visualization tools.



4 QUICKSIGHT AND RDS CONFIGURATION

CONNECTING QUICKSIGHT AND RDS

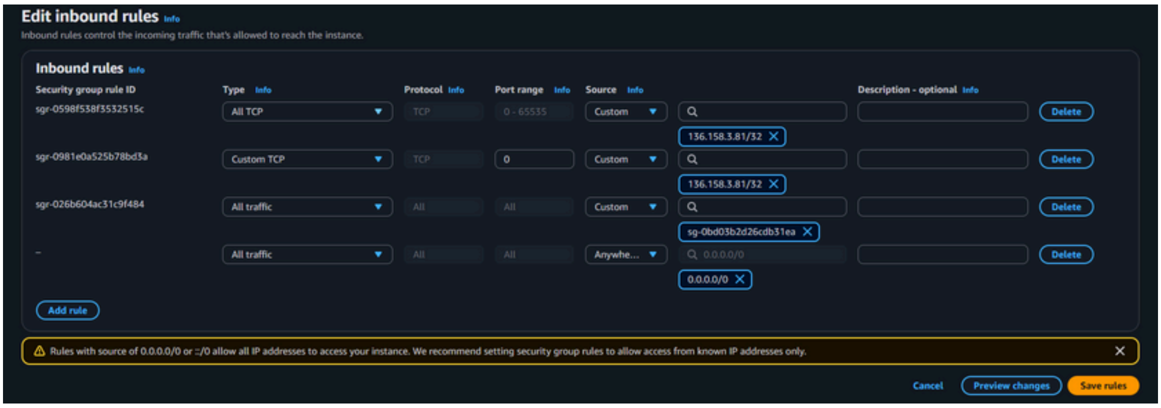
To connect QuickSight to my RDS database, I configured the inbound rules of a dedicated security group. Instead of leaving the database public, which exposed it to all IP addresses, I applied the principle of least privilege by allowing inbound traffic only from QuickSight's specific IP ranges on the database port. This prevented unauthorized access while still enabling QuickSight to retrieve data for visualization.

IMPLEMENTING THE SECURE CONFIGURATION

I then attached this security group to QuickSight during the VPC connection setup. By ensuring the inbound rules matched QuickSight's IP ranges and restricting all other traffic, the database became accessible only to QuickSight within the VPC. This configuration strengthened security by reducing exposure, while still allowing QuickSight to connect seamlessly for secure data analysis.

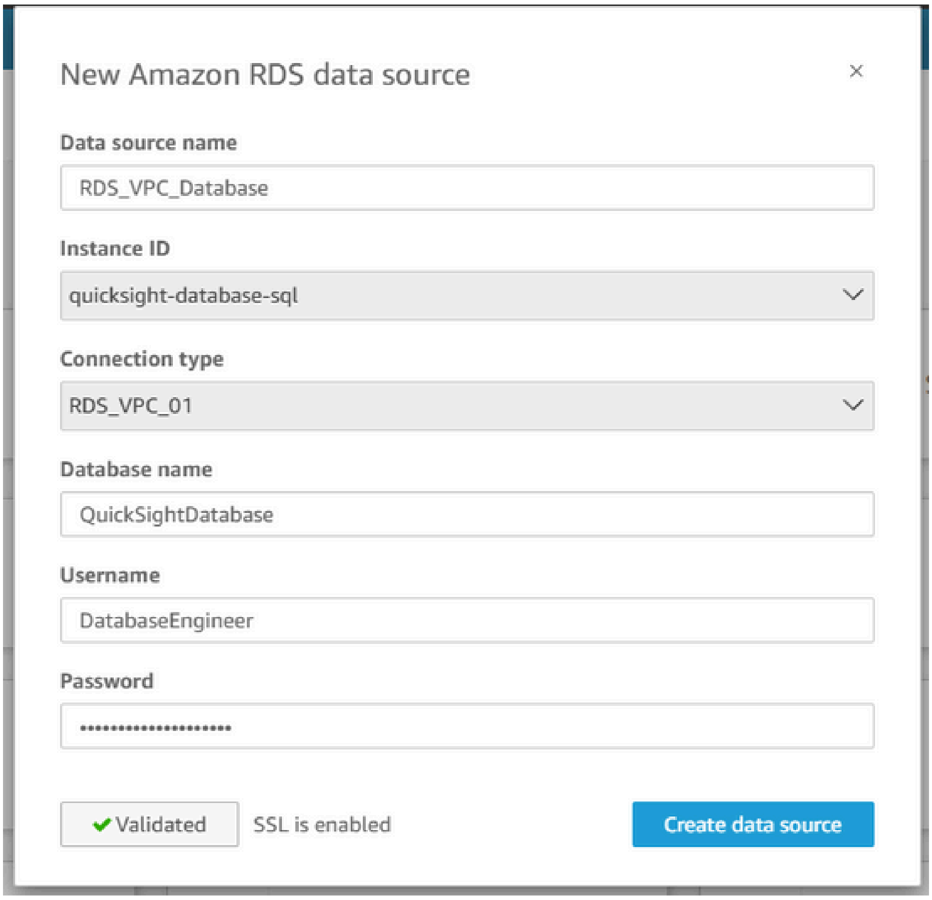
SECURING MY RDS INSTANCE

To secure my Amazon RDS instance, I applied a dedicated security group that strictly manages inbound and outbound traffic. Instead of exposing the database to the public internet, I defined rules that allow access only from trusted sources, such as QuickSight through its VPC connection. This setup ensures the database remains private while still permitting authorized services to connect securely.



ADDING RDS AS A DATA SOURCE FOR QUICKSIGHT

When adding RDS as a data source in QuickSight, I configured it to operate within a private network, ensuring that the database could only be accessed through QuickSight's VPC connection. This approach minimized the risk of public exposure and reduced the chance of unauthorized access, while still allowing secure and seamless integration for data visualization.

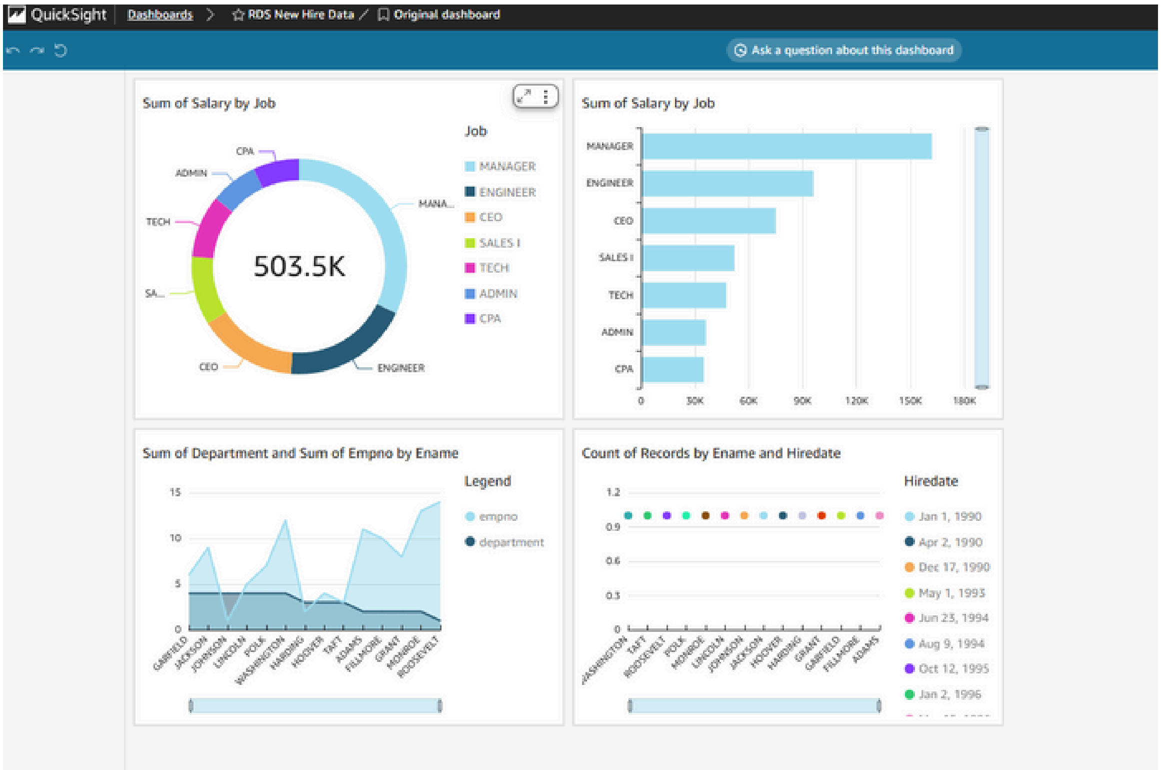


VPC ID	VPC connection ARN	Security Group IDs	DNS resolvers	Status
vpc-03c1cf18b3356fbfc	arn:aws:quicksight:ap-southeast-1:210806...	sg-0f77eb0c6ad0dca87		AVAILABLE

5 VISUALIZING DATA

VISUALIZING DATA IN QUICKSIGHT

After confirming that the RDS database was securely connected, I used the QuickSight console to build visualizations and design a dashboard. This allowed me to transform raw relational data into interactive charts and graphs, such as salary distribution by job role, departmental summaries, and employee hire dates. By leveraging QuickSight’s visualization features, I was able to present the database insights in a clear and accessible way.



CONCLUSION

This project demonstrated how to securely build, manage, and visualize a relational database using Amazon RDS, MySQL Workbench, and Amazon QuickSight. By combining scalable cloud infrastructure with strong security practices, the database was kept private while still being accessible to authorized tools and services. The resulting QuickSight dashboards transformed structured data into meaningful insights, showing the value of integrating AWS services for both data management and visualization. Overall, the project highlights a practical approach to designing cloud-based databases that balance performance, usability, and security.

FUTURE IMPROVEMENTS

For future enhancements, the project could integrate AWS Glue or Lambda to automate data loading and transformations, reducing the need for manual SQL execution. Enabling scheduled refreshes in QuickSight would keep dashboards updated with the latest information in real time. Expanding to a multi-region Aurora deployment could further improve fault tolerance and disaster recovery. Finally, incorporating AWS Secrets Manager for credential management and CloudWatch monitoring for performance insights would strengthen both security and operational efficiency.

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